

BIKIN, Russian Federation

WMO: 318320

Lat: 46.800N

Lon: 134.267E

Elev: 68

StdP: 100.52

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-29.6	-34.1	0.2	-31.6	-32.1	0.2	-29.1	5.7	-13.3	5.0	-13.6	0.4	100	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	29.9	22.1	28.6	21.5	27.1	20.7	23.7	28.0	22.9	26.9	22.0	25.7	1.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.3	26.3	21.5	16.3	25.3	20.6	15.4	24.5	71.9	28.1	68.3	27.0	65.1	25.6	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	5.0	4.3	DB	-34.4	32.7	2.7	1.2	-36.4	33.6	-38.0	34.3	-39.5	35.0	-41.5	35.9	(4)
(5)			WB	-34.5	25.2	2.7	0.8	-36.4	25.7	-38.0	26.2	-39.6	26.6	-41.5	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-19.6	-14.6	-4.8	5.7	13.5	18.7	22.4	20.8	14.4	5.9	-6.1	-16.9		
	DBStd	15.30	4.73	5.20	5.85	4.75	3.96	3.40	2.42	2.70	3.83	4.60	6.45	5.50		
	HDD10.0	3693	917	689	460	148	12	0	0	0	7	143	484	833		
	HDD18.3	5739	1176	923	718	381	157	36	2	9	127	385	734	1092		
	CDD10.0	1271	0	0	0	17	121	262	383	334	138	16	0	0		
	CDD18.3	276	0	0	0	1	7	47	127	85	8	0	0	0		
	CDH23.3	1783	0	0	0	8	95	338	833	460	48	1	0	0		
	CDH26.7	393	0	0	0	1	14	82	216	76	4	0	0	0		
(14) Wind	WSAvg	1.8	1.3	1.5	1.8	2.2	2.2	1.9	1.7	1.7	1.6	1.8	1.9	1.4		
(15) Precipitation	PrecAvg	645	12	9	19	41	61	90	113	125	90	54	27	19		
	PrecMax	847	27	27	69	104	143	258	212	276	165	154	81	56		
	PrecMin	384	0	0	4	11	14	28	14	11	16	0	6	3		
	PrecStd	107	8	7	13	22	26	43	54	57	34	39	16	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.7	3.0	11.7	24.2	28.6	31.4	31.8	30.3	27.1	21.2	11.8	1.9		
		MCWB	-5.0	0.3	5.3	13.1	17.7	21.4	23.6	23.0	19.8	13.8	8.2	-0.2		
	2%	DB	-6.9	0.4	8.6	20.2	26.1	29.0	30.4	28.7	24.7	18.3	7.7	-2.1		
		MCWB	-8.3	-2.1	3.3	11.3	16.3	20.5	22.7	22.2	18.5	12.0	4.3	-3.1		
	5%	DB	-9.6	-2.9	6.3	16.9	23.5	27.0	29.1	27.2	22.8	15.8	5.4	-5.3		
		MCWB	-10.7	-4.9	1.9	9.2	14.8	19.4	22.2	21.4	17.3	10.7	2.7	-6.4		
	10%	DB	-11.8	-5.6	4.1	14.4	21.3	25.1	27.6	25.9	21.1	13.7	3.0	-8.3		
		MCWB	-12.7	-7.3	0.6	7.7	13.6	18.7	21.7	20.8	16.3	9.2	0.9	-9.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	1.0	6.0	13.7	18.7	23.6	25.2	24.3	21.1	15.1	7.8	0.3		
		MCDB	-3.9	2.3	9.6	22.8	26.9	29.4	29.7	28.2	25.2	19.7	12.2	1.8		
	2%	WB	-8.3	-2.2	3.8	11.4	17.0	21.7	24.2	23.3	19.5	12.9	4.6	-3.0		
		MCDB	-7.0	0.3	8.0	18.9	24.4	27.1	28.5	27.1	23.3	16.6	7.1	-2.1		
	5%	WB	-10.6	-4.8	2.3	9.7	15.7	20.5	23.3	22.5	18.2	11.5	2.8	-6.3		
		MCDB	-9.7	-2.8	5.5	16.0	22.1	25.4	27.6	26.0	21.7	15.2	5.1	-5.3		
	10%	WB	-12.6	-7.3	0.8	8.1	14.3	19.4	22.5	21.7	17.0	9.7	1.1	-9.2		
		MCDB	-11.7	-5.7	4.0	13.6	19.8	23.9	26.4	24.9	20.2	13.1	2.8	-8.3		
(35) Mean Daily Temperature Range	MDBR		11.9	13.5	11.5	11.5	11.0	9.5	8.6	8.5	11.0	10.7	9.4	10.4		
	5% DB	MCDBR	13.0	13.0	12.1	15.9	15.1	12.5	11.4	10.5	12.7	13.9	10.6	10.7		
		MCWBR	11.9	11.0	7.8	8.8	7.5	5.6	4.9	4.9	6.7	8.3	7.8	10.0		
	5% WB	MCDBR	12.7	12.4	10.9	14.6	13.1	10.8	9.6	8.4	10.5	11.9	9.5	10.6		
		MCWBR	11.6	10.6	7.4	8.3	7.0	5.4	4.8	4.4	6.1	7.6	7.3	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.346	0.428	0.537	0.570	0.536	0.510	0.468	0.422	0.440	0.384	0.320		
	taud		2.089	1.953	1.811	1.746	1.751	1.928	2.051	2.168	2.203	1.990	2.029	2.056		
	Ebn at Noon		752	785	768	721	717	748	763	780	780	671	632	669		
	Edh at Noon		93	133	180	212	219	184	161	138	121	129	101	87		
(44) All-Sky Solar Radiation	RadAvg		2.01	3.18	4.34	4.57	5.18	5.54	5.16	4.61	3.91	2.66	1.77	1.53		
	RadStd		0.12	0.15	0.34	0.46	0.46	0.68	0.47	0.29	0.34	0.20	0.15	0.10		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page